COGNITEK FRONTEND: THE "DON'T PANIC" GUIDE

Obviously, u going to encounter errors during package and dependency installation, don’t give up. Copy paste the errors and fix them.
FIGHT THROUGH THE END.

- **Location:** `/frontend/cognitek_web`
- **The Goal:** Getting the UI live without calling your mom.

THE TECH STACK (Why we chose this pain)

Tool	Why it’s here
React	Component-based architecture. Great for reusable UI and not losing your mind when things scale.
Vite	Lightning-fast dev server. Because life is too short to wait for Webpack to compile.
Capacitor	The "Native Bridge." We used this instead of Flutter so we could stay in Web-land while still stealing the phone's microphone.
Tailwind CSS	For when you don't want to write 4,000 lines of CSS just to center a `div`.
Supabase	The brain. Handles the login, the data, and the secrets.

🗁 PROJECT STRUCTURE (Where the bodies are buried)

- `src/components/` → Reusable UI (Recorders, Forms, Boards).
- `src/pages/` → The actual views (Dashboard, Study, Record).
- `src/context/` → The "Global State" (Auth, Themes, Auto-Record logic).
- `src/hooks/` → Custom logic (PWA stuff, WakeLocks).
- `src/api/` → The backend handshake logic.
- `src/lib/` → Configuration for things like the Supabase client.

💻 LOCAL DEVELOPMENT SETUP

1. Prerequisites

- **Node.js:** Version 18 or higher (If you have v14, just stop now and update).
- **Package Manager:** npm or yarn.
- **Hardware Access:** Android Studio (Windows/Linux) or Xcode (Mac) for mobile testing.

2. Installation

Bash

```bash
# Navigate to the correct folder
cd Frontend/cognitek_web

# Install the dependencies (Go get a coffee, this takes a minute)
npm install
```

🏃 RUNNING THE APPLICATION

A. The Web Version (Most of your work)

Bash

```bash
npm run dev
```

- Opens at `http://localhost:5173`.
- Includes **Hot Module Replacement** (HMR)—it updates the second you hit Save.

B. The Mobile Bridge (The Capacitor Flow)

We implemented the **Capacitor method**, not a native Flutter build. You need Android Studio for this.

Step 1: Sync the code
Bash

```bash
npx cap sync android
```

Push your latest web changes into the Android project.

Step 2: Open the Native Project
Bash

```bash
npx cap open android
```

This launches Android Studio. From there, you can build the APK or run it on an emulator.

☁️ CLOUD & DEPLOYMENT

- **Backend:** Hosted on **Render** (`https://cognitek-backend.onrender.com/api`).
- **Auth/Data:** Managed by **Supabase**. If the login fails, it's probably because the Supabase keys in your `.env` are wrong.
- **PWA Status:** This is a Progressive Web App. It has a **Service Worker** for offline use.
  - *Test it:* `npm run build` → `npm run preview`. Check the 'Application' tab in DevTools.

🚨 TROUBLESHOOTING

(The "I broke it" section)

1. **"Recording Fails":** It’s a 99% chance it's a **Permission Issue**. Check if the app actually has microphone access in Android settings.

2. **"Build Errors":** The nuclear option:

   Bash

   ```
   rm -rf node_modules && npm install
   ```

3. **"API Connection Issues":** Check the URL in `src/api/api.js`. If the backend is "sleeping" on Render, the first request might take 30 seconds to wake up.

🎓 HANDOVER FOR NEW DEVELOPERS

1. **Clone it.**
2. **`npm install`** (Prepare for warnings, ignore the ones that aren't red).
3. **`npm run dev`** to see the magic.
4. If you're doing mobile, get **Android Studio** ready.
5. Read the **Supabase docs**. Seriously. Don't just wing it.

Final Implementation Note: We went with a hybrid web/native solution to save time and leverage web skills. It's efficient, it's fast, and it works—as long as you don't mess up the Capacitor sync.

Now go build something cool and don't break the codebase. Good luck.

THE END

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